

What This Study Can and Cannot Tell You

Applying the trajectories data to a PhD-or-build decision

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Read this before any number below

You collected this data. You already know the caveat, so this is the short version: **the median is a duration, not a probability**. It says how long the road ran for people who reached the end. It says nothing about how many set out.

That has a specific consequence for the decision in front of you. This study **cannot** tell you whether to do a PhD or start a company, because it never sampled anyone who tried either and failed. Both arms of your decision are represented here only by their winners. Any sentence of the form “the data says path A beats path B” is unsupported, and I am not going to write one.

What the data *can* do is describe the shape and timing of careers that worked. That is genuinely useful for planning, and useless for forecasting.

1 The four numbers that bear on your decision

Quantity	Median	Spread
Age at first hit	33.0	IQR 30–39, $n=105$
if the hit landed post-1995	32.0	IQR 28.8–36.0
Years from age 18 to first hit	15.0	IQR 12–21
Years from starting the venture that worked	4.2	IQR 3.0–7.2, $n=118$

You are 22. The median person in this dataset who eventually crossed \$10M constant did it at 33. That is eleven years from where you stand, and **both of your options fit inside it**. A five-year PhD ending at 27 leaves you six years to the median. Starting a company now leaves you eleven.

Nothing about the timing forces the choice. That is the first useful finding: on this evidence, the decision is not a race against a closing window.

2 The reframe: the long pole is not the company

Put two of those numbers next to each other. From age 18 to the hit is a median of 15 years. From *starting the venture that worked* to the hit is 4.2.

Roughly eleven of those fifteen years happen before the winning venture starts.

The company is the short part. The long part is whatever you were doing beforehand — and in this sample, that period is not idle. It is where domain knowledge, capital, credibility and the earlier failed attempts accumulate. The study’s own rule that a serial founder is dated at their **first** company to cross, not their famous one, means those earlier attempts are inside the eleven years, not excluded from them.

So the real question is not “PhD or company.” It is:

Which pre-venture decade puts me in a position where a venture takes four years instead of never working at all?

A PhD is one way to spend part of that decade. So is building and failing twice. So is working for someone competent in an industry you intend to attack. This data cannot rank them. But it does say the pre-venture period is where the time actually goes, which means choosing it casually is the expensive error — not choosing it slowly.

3 Where the two paths diverge in the data

The one place this dataset speaks directly to research-versus-commerce is the venture clock by field, and it is stark:

Field	<i>n</i>	Median years from venture start to hit
Software / internet	54	4.0
Hardware / deep tech	33	4.0
Science / research	10	14.0

Read this carefully, because the obvious reading is wrong. These are not the same test. For the commercial fields, the hit is \$10M of revenue in constant dollars — a good business, achieved by thousands of people per decade. For science, the hit criterion is a **major prize**: a Nobel, a Fields Medal, a Turing Award, a Breakthrough Prize. That is a handful of people per year on the planet.

So the honest statement is: *reaching the very top of academic science takes about fourteen years from starting the work, while building a \$10M business takes about four from starting the company.* Those are not comparable outcomes and the ratio between them is not a return on investment. If anything the comparison flatters research, because the science rows are sampled at a far more extreme threshold and still show up in a dataset this small.

What you can take from it: **academic recognition is a slow clock even for its winners**, and the sample contains nobody who got there quickly. Also note $n=10$. Treat that 14.0 as an impression, not a measurement.

Also relevant: software has the youngest median age at hit in the study, 30.0, against 34.5 for hardware and 36.0 for healthcare. Fields differ in when they pay, by five or six years.

4 The factor this data cannot see, and it may dominate

Your decision has a structural asymmetry the dataset has no column for: **immigration status**.

Nobody in this sample was selected on visa constraints. Several were immigrants — but the study recorded where their career ran, never what permitted it to run there. That is a real hole for you specifically, and it cuts one way:

- A US PhD is not only training. It is roughly five more years of lawful presence, an assistantship that pays, OPT afterwards, repeated H-1B lottery entries, and eligibility arguments for EB-1B or an EB-2 national interest waiver that a bachelor's alone does not support.
- Going home to Busan after the bachelor's to build is not a neutral pause. It substantially closes the US route for years, and reopening it later is harder than not closing it.
- A US business started on F-1 has narrow lawful room to operate before OPT, and that constraint is on the timeline whether or not the business is good.

I want to be precise about the logic here: **this is not an argument that the PhD is better.** It is an argument that the two options are not symmetric in what they foreclose. One of them keeps more branches open at roughly equal cost in years. When a decision is genuinely close on the merits — and on this evidence it is — optionality is the tiebreaker that survives being wrong.

The camera-import business is the interesting third case precisely because it is the one that could plausibly run from either country, which makes it the option least coupled to this asymmetry. That is worth more than it sounds.

5 What would actually change my read

Rather than a recommendation, here is what I would want you to answer, because each one moves the decision more than any number in this study:

1. **Do you have a specific thing to build, or a category?** Everyone in this dataset crossed with a particular product for a particular buyer. “Start a business” is not a plan and the four-year venture clock does not apply to it. If the answer is a category, the eleven pre-venture years have not started being spent well yet.
2. **Is the PhD in something adjacent to what you would build?** Physics to deep tech or instrumentation keeps both paths alive at once. Physics to academic physics is a commitment to the 14-year clock and its far narrower gate.
3. **Can you test the business inside the next twelve months without giving anything up?** The camera-import idea is unusually testable — it has real customers, real margins, and a real cash cycle you can observe. A quarter of genuine trading tells you more than this entire study can.
4. **What happens if you are still pre-hit at 33?** The median says half of the winners were. Whichever path leaves you employable, solvent and in a country you chose at 33 is the one that survives the median outcome rather than the lucky one.

6 The honest summary

The data supports exactly three statements about your decision:

1. **You are not late, and there is no closing window.** The median winner crossed at 33; the middle half crossed between 30 and 39.
2. **The venture is the short part.** About eleven of the fifteen years from age 18 happen before the winning company starts. Choose the pre-venture decade deliberately; that is where the time actually goes.

3. **Top-of-field academic recognition is a slow clock even for winners**, roughly fourteen years from starting the work, on thin evidence and at a far more extreme threshold than the commercial rows.

Everything past that is not in the data. The visa asymmetry is real and points toward keeping US presence. The strongest argument for building now is having something specific to build, and this study cannot tell you whether you do.

One last time, because it is the misreading this dataset invites most: nine years, or thirty-three, is how long it took the people who made it. It is not your odds, and it is not a schedule. The people who spent the same decade and got nothing are not in this file — not because they were rare, but because the study was built in a way that cannot see them.